

Frank Liu

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EDUCATION

Princeton University

Bachelor of Science in Engineering (B.S.E) in Computer Science

Princeton, NJ

Expected Graduation: May 2026

RELEVANT COURSEWORK

Object-Oriented Programming, Algorithms and Data Structures, Programming Systems, Discrete Mathematics, Introduction to Machine Learning, Linear Algebra, Intro to Software Engineering, Principles of Computer System Design, Data Science, Building a Startup

WORK EXPERIENCE

Software Development Engineer Intern

Amazon Web Services, Kiro Team

May 2025 - August 2025

Seattle, WA

- Contributed core onboarding and migration features that shipped with the launch of Kiro, AWS's agentic AI IDE, helping drive a highly successful public release
- Reduced developer repository onboarding time by automating multi-provider configuration imports from major AI assistants (Cursor, Windsurf, Cline, etc) using TypeScript, React, and Node.js
- Boosted adoption and reliability by delivering scalable VS Code extension features and ensuring high test coverage with Mocha, Chai, and Sinon through TDD

Teaching Assistant

COS226 & COS217 (Data Structures/Algorithms & Programming Systems)

January 2024 - Present

Princeton, NJ

- Chosen based on top performance in the courses; hold lab hours twice a week for 400+ students
- Assist students with debugging and optimizing code for complex data structures, algorithms, and systems programming assignments (Java and C)

Product Design Intern

22x36 - Chinese Language Learning App

December 2023 - January 2024

Remote

- Researched rewards and progress-tracking gamification strategies to enhance user engagement with interactive practice features
- Explored AI tools like ChatGPT, Dall-E 3, and Anthropic to develop innovative and interactive language learning content aligned with the Hanyu Shuiping Kaoshi
- Provided insights and feedback on user experience during the initial stages of website design

PROJECTS

Digital Wardrobe App | *Swift/SwiftUI*

- Developed a mobile app that allows users to upload photos of clothing items, plan outfits, and organize them into themed collections for easy outfit management, designed using Figma

Word Hunt Solver and Learner | *React, Flask, Python*

- Leveraged React and Flask to create a web app that solves Apple's GamePigeon Word Hunt game using Depth-First Search (DFS) to find all possible words from user input
- Expanded vocabulary learning opportunities by outputting word paths and definitions using PyDictionary module

TigerSpot | *Python, Flask, JavaScript, PostgreSQL*

- Developed a full-stack web application using HTML/CSS, JavaScript, Flask, and PostgreSQL, creating a location guessing game for Princeton students and faculty to learn more about various locations and landmarks on campus
- Promoted campus engagement for 200+ users by designing a PostgreSQL relational database to manage leaderboards, track user ranks and points, and support an asynchronous versus mode where players compete to score points by guessing the locations of five campus images
- Ensured smooth handling of game visuals and performance using Render for deployment and Cloudinary for image storage

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Assembly, Go, R, Swift

Technologies: Flask, React, SQLite, PostgreSQL, NumPy, Node.JS

Tools: Git, REST, Agile, VSCode, IntelliJ, Kiro